

10 Audience Questions and Answers

The following questions represent potential stakeholder inquiries based on the analysis.

Question 1:

How accurately can the model predict customer churn?

Answer:

The model predicts customer churn with high accuracy. The Random Forest model achieved approximately 95.5% accuracy and a ROC-AUC score of 0.987. This indicates that the model performs very well in distinguishing between existing and churned customers.

Question 2:

What are the most important factors influencing customer churn?

Answer:

The two most important factors are Total_Trans_Amt (Total Transaction Amount) and Total_Trans_Ct (Total Transaction Count) over the past twelve months. These are behavioral features, showing that customer activity and spending patterns play a key role in churn. Demographic features did not show strong influence.

Question 3:

How can banks use this model to improve customer retention strategies?

Answer:

Banks can use this model to monitor customer transaction behavior. By tracking total spending and number of transactions, they can identify declining activity. If these metrics drop below certain levels, banks can take proactive steps such as offering cashback, promotions, or incentives to encourage continued card usage.

Question 4:

Is this data collected from US or International banks?

Answer:

The dataset does not explicitly specify the source. The values are shown in US dollars, which suggests it may be from US-based data, but this cannot be confirmed. Therefore, this is a limitation of the dataset.

Question 5:

Can different regions or countries influence spending habits and churn rate?

Answer:

Yes, regional differences can influence spending behavior and churn. For example, in Europe, customers often prefer debit cards, while in North America, credit card usage is more common due to credit history building and reward programs. These differences can affect churn patterns.

Question 6:

How does class imbalance affect the model's performance?

Answer:

Class imbalance means that one group is much larger than the other. In this dataset, existing customers significantly outnumber churned customers. This can make it harder for the model to correctly learn patterns for the smaller group, potentially reducing prediction accuracy for churned customers.

Question 7:

Why did Random Forest perform better than Logistic Regression?

Answer:

Random Forest performed better because it can capture more complex relationships in the data. Logistic Regression is a simpler model and may not perform as well when relationships between variables are not linear. Random Forest is also more robust when working with imbalanced datasets.

Question 8:

Can this model be applied to other banks or datasets?

Answer:

Yes, this model can be applied to other banking products such as auto loans, mortgages, or student loans. It can also be used in other industries like telecommunications or insurance, where customer churn is a common problem.

Question 9:

How often should the model be retrained?

Answer:

The model should be retrained regularly, depending on changes in customer behavior. In stable conditions, retraining every few months may be sufficient. However, during periods of economic change, such as shifts in interest rates or employment, more frequent retraining may be needed.

Question 10:

What additional data could improve the model?

Answer:

Additional data such as customer service interactions, complaints, and feedback could improve the model. Information about promotions, competitor offers, and customer satisfaction would also help provide deeper insights into why customers decide to leave.